Partial Gaussian splitting in quadratic Wasserstein distance

Sam Vaseghi[1*]

[1]Wisehouse Science Communication Lab, Wisehouse AB, Hässleholm, Sweden.

Corresponding author(s). E-mail(s):
sam.vaseghi@wisehouse-publishing.com;

Abstract

Near equality in the Poincaré inequality suggests Gaussian structure, but a partial splitting estimate must also control dependence on the remaining coordinates. Let $\boldsymbol{\mu = e^{-V}\,dx}$ be a smooth probability measure on $\mathbb{R}^{\boldsymbol{n}}$ with $\boldsymbol{\nabla^2 V \succeq I_n}$. For $\boldsymbol{1 \le k \le n}$, suppose that $\boldsymbol{u_1, \dots, u_k}$ are centered, variance-one Sobolev functions whose gradients are orthogonal in $\boldsymbol{L^2(\mu)}$, and let $\boldsymbol{D = \sum_{i=1}^{k} (\int |\nabla u_i|^2 \, d\mu - 1)}$. We prove that there is a $\boldsymbol{k}$-dimensional subspace $\boldsymbol{E}$ for which the squared quadratic Wasserstein distance from $\boldsymbol{\mu}$ to the product of a standard Gaussian on $\boldsymbol{E}$, translated by the projected mean, and the actual marginal of $\boldsymbol{\mu}$ on $\boldsymbol{E^{\perp}}$ is at most $\boldsymbol{15D}$. The estimate has no ambient-dimension factor and requires no smallness assumption on $\boldsymbol{D}$. The proof combines the Caffarelli contraction and Gaussian near-affinity estimates with a simultaneous selection of $\boldsymbol{k}$ directions and an explicit coupling that preserves the complementary marginal. When $\boldsymbol{k = n}$, this gives full Gaussian approximation, while zero total deficit gives an exact independent Gaussian factor.

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ORCID: https://orcid.org/0009-0006-4105-1949.

1 The joint splitting estimate

For a probability measure whose potential satisfies $\nabla^2 V \succeq I_n$, the Poincaré inequality bounds the variance of a function by its Dirichlet energy. When a nonconstant function attains equality, a Gaussian direction appears in the measure. Quantitative versions ask whether several functions close to equality force an approximately independent Gaussian factor. Courtade and Fathi [1] proved such a conclusion in W_1, following the rigidity and stability analysis of De Philippis and Figalli [2]. Remark 1.1 of [1] distinguishes full Gaussian approximation from the partial-factor case and raises the corresponding W_2 question.

The distinction concerns the joint distribution. A projection can be close to Gaussian while its dependence on the remaining coordinates persists. We therefore compare the original law with a product whose second factor is its own complementary marginal, so that the transport cost measures both the Gaussian approximation and the removal of dependence.

Let γ_n denote standard Gaussian measure on $\mathbb{R}^n$. For a subspace E and $b \in E$, write $\gamma_{b,E}$ for the Gaussian measure on E with mean b and identity covariance on E. Let P_E be the orthogonal projection onto E. We identify a product measure on $E \times E^\perp$ with its image in $\mathbb{R}^n$ under addition. For probability measures with finite second moments, our convention is

$$W_2(\rho,\eta)^2 = \inf_{\pi\in\Pi(\rho,\eta)} \int_{\mathbb{R}^n\times\mathbb{R}^n} |x-y|^2 \, d\pi(x,y),$$

where $\Pi(\rho,\eta)$ is the set of couplings of ρ and η. We write $H^1(\mu)$ for the space of functions in $L^2(\mu)$ whose weak gradients also belong to $L^2(\mu)$. The notation $S_\#\mu$ denotes the law of $S(X)$ when X has law μ.

Theorem 1.1 *Let $\mu = e^{-V}\,dx$ be a probability measure on $\mathbb{R}^n$, where $V \in C^\infty(\mathbb{R}^n)$ and $\nabla^2 V \succeq I_n$. Let $1 \le k \le n$ and suppose that $u_i \in H^1(\mu)$ satisfy*

$$\int u_i \, d\mu = 0, \qquad \int u_i^2 \, d\mu = 1, \qquad \int \nabla u_i \cdot \nabla u_j \, d\mu = 0 \quad (i \neq j). \tag{1.1}$$

Set

$$q_i = \int |\nabla u_i|^2 \, d\mu, \qquad D = \sum_{i=1}^k (q_i - 1), \qquad m = \int x \, d\mu.$$

There is a k-dimensional subspace $E \subseteq \mathbb{R}^n$ such that

$$W_2\big(\mu, \gamma_{P_E m,E} \otimes (P_{E^\perp})_\#\mu\big)^2 \le 15D. \tag{1.2}$$

In particular, if $q_i \le 1+\varepsilon$ for every i, the right-hand side is at most $15k\varepsilon$.

The Poincaré inequality gives $q_i \ge 1$, so $D \ge 0$. The Gaussian factor in (1.2) is independent of the actual complementary marginal and is centered at the projected mean of μ. The hypothesis is orthogonality of the gradients after integration; pairwise orthogonality of the functions themselves is unnecessary. The result includes $k = n$,

when the target is the full Gaussian with mean m, and $D = 0$, when it gives exact factorization. The constant 15 is not asserted to be optimal.

The proof uses Caffarelli's contraction theorem [3]. The transport-to-product strategy, including replacement of an auxiliary factor by the actual complementary marginal, already appears in [2, Section 3]; the Gaussian near-affinity argument is also established in [2, Section 4] and [1]. Here a Gaussian-input resampling coupling controls the joint squared transport cost, while a matrix argument selects all k directions at once. The coupling bounds the squared transport cost by three times a derivative defect, and the direction-selection argument bounds that defect by $5D$. Together they give (1.2) without inversion of a Gram matrix or a small-deficit threshold.

Other recent stability results concern related quantities. Courtade and Fathi's analysis of Klartag's improved Lichnerowicz inequality [4] gives one-factor W_1 stability with an additional directional Fisher-information parameter. Fathi and Violo, in their 2026 preprint *Quantitative stability in distribution for the Sobolev inequality under curvature dimension condition* (arXiv:2607.05304v1, Theorem 5.1(ii)), obtain W_2 approximation for the scalar distribution of a normalized near-extremizer on an $\mathrm{RCD}(1,\infty)$ space. Estimate (1.2) concerns the original Euclidean joint law, including its independence from the remaining coordinates.

2 A coupling that preserves the complementary marginal

Write $T = \nabla\Phi$ for the Brenier transport from γ_n to μ. Caffarelli's theorem gives a 1-Lipschitz representative of T whose derivative satisfies

$$A(x) := DT(x) = A(x)^T, \qquad 0 \preceq A(x) \preceq I_n \quad \text{for almost every } x. \tag{2.1}$$

All expectations in the proof below are with respect to standard Gaussian inputs, unless another law is indicated. Uniform convexity ensures finite second moments, so that $m = \mathbb{E}T$ is well defined. We use $\|M\|_{\mathrm{HS}}^2 = \mathrm{tr}(M^T M)$ for the squared Hilbert–Schmidt norm.

For a fixed subspace E, the quantity $\mathbb{E}\|(I_n - A)P_E\|_{\mathrm{HS}}^2$ measures how closely the transport preserves the Gaussian directions in E. The next proposition converts this derivative estimate into a coupling with the required product law.

Proposition 2.1 *For every subspace $E \subseteq \mathbb{R}^n$, with $P = P_E$ and $Q = I_n - P$, one has*

$$W_2\big(\mu, \gamma_{Pm,E} \otimes Q_{\#}\mu\big)^2 \le 3\mathbb{E}\|(I_n - A)P\|_{\mathrm{HS}}^2. \tag{2.2}$$

Proof Take independent $Z, Z' \sim \gamma_{0,E}$ and $W \sim \gamma_{0,E^\perp}$, and define

$$X = T(Z + W), \qquad Y = Z + Pm + QT(Z' + W). \tag{2.3}$$

Both $Z + W$ and $Z' + W$ are standard Gaussian on $\mathbb{R}^n$, so X has law μ and $QT(Z' + W)$ has law $Q_{\#}\mu$. Since Z is independent of (Z', W), the two components of Y are independent. Thus Y has precisely the product law in (2.2).

Orthogonality of E and $E^\perp$ splits the squared cost into

$$\begin{aligned}\mathbb{E}|X-Y|^2 &= \mathbb{E}|PT(Z+W)-Z-Pm|^2 \\ &\quad + \mathbb{E}|QT(Z+W)-QT(Z'+W)|^2. \end{aligned} \tag{2.4}$$

For the first term, apply the Gaussian Poincaré inequality componentwise to $P(T(x)-x)$. Its mean is Pm, and symmetry of A gives

$$\mathbb{E}|PT(Z+W)-Z-Pm|^2 \le \mathbb{E}\|P(A-I_n)\|_{\mathrm{HS}}^2 = \mathbb{E}\|(A-I_n)P\|_{\mathrm{HS}}^2. \tag{2.5}$$

For the second term, condition on W. The two vectors are conditionally independent copies. If the variance of a vector denotes the sum of its coordinate variances, the independent-copy identity and Gaussian Poincaré inequality on E yield

$$\begin{aligned}\mathbb{E}|QT(Z+W)-QT(Z'+W)|^2 &= 2\mathbb{E}_W \operatorname{Var}_Z\big(QT(Z+W)\big) \\ &\le 2\mathbb{E}\|QAP\|_{\mathrm{HS}}^2 \\ &\le 2\mathbb{E}\|(A-I_n)P\|_{\mathrm{HS}}^2. \end{aligned} \tag{2.6}$$

The last inequality follows from $QAP = Q(A-I_n)P$. Lipschitz continuity of T justifies both Poincaré applications in the Sobolev sense. Combining (2.4)–(2.6) proves the proposition. □

The shared variable W couples the original and target vectors, while the fresh variable Z' removes dependence between the two target components. This is why the construction preserves the complete law of the complement without requiring that law to be Gaussian or to have a small dimension.

3 Selection of the Gaussian directions

Proposition 2.1 reduces the theorem to finding k orthogonal directions with small total derivative defect. We first pull the near-extremizing functions back to Gaussian space and use the separation between the linear and higher Hermite components to find candidate directions. We then select an orthonormal frame through a trace estimate that remains valid when the candidate vectors are linearly dependent.

3.1 Pullback and near-affinity

For each i, put

$$v_i = u_i \circ T, \qquad g_i = (\nabla u_i)\circ T, \qquad f_i = \nabla v_i = Ag_i, \qquad a_i = \mathbb{E} f_i.$$

These definitions extend to $u_i \in H^1(\mu)$. Indeed, choose compactly supported smooth approximants in weighted $H^1(\mu)$. Composition with T preserves their L^2 distances because $T_{\#}\gamma_n = \mu$, and the contraction property gives

$$\|\nabla((u-\widetilde{u})\circ T)\|_{L^2(\gamma_n)} \le \|\nabla(u-\widetilde{u})\|_{L^2(\mu)}$$

for smooth $u, \widetilde{u}$. Closure of the weak gradient proves the stated chain rule. Smooth positive densities provide the required weighted Sobolev approximation; no normalization of the approximants is needed.

The function v_i has mean zero and variance one. Define

$$r_i = \mathbb{E}|f_i|^2, \qquad s_i = r_i - 1, \qquad \delta_i = q_i - r_i.$$

Gaussian Poincaré and (2.1) imply

$$1 \le r_i \le q_i, \qquad s_i, \delta_i \ge 0, \qquad s_i + \delta_i = q_i - 1. \tag{3.1}$$

The Hermite decomposition has the form $v_i = a_i \cdot x + z_i$, where z_i is orthogonal to constants and linear functions. Gaussian integration by parts identifies the linear coefficient with $\mathbb{E}\nabla v_i = a_i$. The spectral inequality on Hermite degrees at least two gives

$$\mathbb{E}|\nabla z_i|^2 \ge 2\mathbb{E}z_i^2, \qquad s_i = \mathbb{E}|\nabla z_i|^2 - \mathbb{E}z_i^2.$$

Consequently,

$$\mathbb{E}|f_i - a_i|^2 = \mathbb{E}|\nabla z_i|^2 \le 2s_i. \tag{3.2}$$

The loss under contraction is controlled separately. Since $(I_n - A)^2 \preceq I_n - A^2$,

$$\mathbb{E}|(I_n - A)g_i|^2 \le \mathbb{E}|g_i|^2 - \mathbb{E}|Ag_i|^2 = \delta_i. \tag{3.3}$$

Using $f_i = Ag_i$, the commutation of A with $I_n - A$, and their operator-norm bounds, we obtain

$$\begin{aligned}\|(I_n - A)a_i\|_{L^2} &\le \|(I_n - A)(a_i - f_i)\|_{L^2} + \|(I_n - A)Ag_i\|_{L^2} \\ &\le \sqrt{2s_i} + \sqrt{\delta_i}.\end{aligned}$$

The inequality $(\sqrt{2s} + \sqrt{\delta})^2 \le 3(s + \delta)$ therefore gives

$$\mathbb{E}|(I_n - A)a_i|^2 \le 3(q_i - 1). \tag{3.4}$$

3.2 An aggregate frame estimate

The vectors a_i have small total defect by (3.4), but they need not be orthonormal or even linearly independent. To control the cost of replacing them by k orthogonal directions, define the $k \times k$ matrices

$$G_{ij} = \mathbb{E}(g_i \cdot g_j), \qquad F_{ij} = \mathbb{E}(f_i \cdot f_j), \qquad H_{ij} = a_i \cdot a_j.$$

The hypotheses and (3.1) give $G = \operatorname{diag}(q_1, \dots, q_k) \succeq I_k$. Applying $I_n - A^2 \succeq 0$ to linear combinations of the g_i shows that $G - F \succeq 0$, while $F - H \succeq 0$ is a covariance matrix. Hence

$$L := G - H \succeq 0, \qquad \operatorname{tr} L = \sum_i \bigl(\delta_i + \mathbb{E}|f_i - a_i|^2\bigr) \le \sum_i (\delta_i + 2s_i) \le 2D. \tag{3.5}$$

Let $\lambda_1, \dots, \lambda_k \ge 0$ be the eigenvalues of H, and let R be its spectral projection onto eigenvalues below one. Writing $(t)_+ = \max(t, 0)$, we have

$$\begin{aligned}\sum_j (1-\lambda_j)_+ &= \operatorname{tr}\big(R(I_k - H)\big) \\ &= \operatorname{tr}\big(R(I_k - G + L)\big) \le \operatorname{tr}(RL) \le \operatorname{tr} L \le 2D. \end{aligned} \tag{3.6}$$

The inequalities use $G \succeq I_k$, $L \succeq 0$ and $0 \preceq R \preceq I_k$; they do not require these matrices to commute.

Let M be the $n \times k$ matrix with columns a_i, so that $H = M^T M$. The positive eigenvalues of MM^T are therefore precisely the positive eigenvalues of H. Choose orthonormal vectors $e_1, \dots, e_k$ so that

$$MM^T = \sum_{j=1}^{k} \lambda_j e_j e_j^T.$$

For positive eigenvalues these are left singular directions of M. When its rank is below k, complete those directions by arbitrary orthonormal vectors, which is possible because $k \le n$.

Set $B = \mathbb{E}(I_n - A)^2$. By (2.1), $0 \preceq B \preceq I_n$, so $d_j = e_j^T B e_j \in [0, 1]$. The scalar inequality $d_j \le \lambda_j d_j + (1-\lambda_j)_+$ now gives

$$\begin{aligned}\sum_{j=1}^{k} d_j &\le \sum_{j=1}^{k} \lambda_j d_j + \sum_{j=1}^{k} (1-\lambda_j)_+ \\ &= \operatorname{tr}(BMM^T) + \sum_{j=1}^{k} (1-\lambda_j)_+ \\ &= \sum_{i=1}^{k} \mathbb{E}|(I_n - A)a_i|^2 + \sum_{j=1}^{k} (1-\lambda_j)_+ \le 5D. \end{aligned} \tag{3.7}$$

The eigenvalue deficit pays for any missing directions, including zero eigenvalues, without dividing by a singular value.

Proof of Theorem 1.1 Take $E = \operatorname{span}(e_1, \dots, e_k)$ from (3.7). Then

$$\mathbb{E}\|(I_n - A)P_E\|_{\mathrm{HS}}^2 = \sum_{j=1}^{k} \mathbb{E}|(I_n - A)e_j|^2 = \sum_{j=1}^{k} d_j \le 5D.$$

Proposition 2.1 yields (1.2). If $q_i \le 1 + \varepsilon$, then $D \le k\varepsilon$, which proves the final assertion. □

Remark 3.1 There is also a universal bound $W_2(\mu, \gamma_{P_E m, E} \otimes (P_{E^\perp})_{\#}\mu)^2 \le 2k$ for every k-dimensional E. To see this, take $X \sim \mu$ and an independent $Z \sim \gamma_{0,E}$, and couple X to $Z + P_E m + P_{E^\perp} X$. Its squared cost is $\operatorname{tr}(P_E \operatorname{Cov}(\mu)) + k \le 2k$, since the Poincaré inequality for linear functions gives $\operatorname{Cov}(\mu) \preceq I_n$. Thus the theorem improves this elementary bound whenever $15D < 2k$, while remaining valid for every finite deficit.

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